JEE APEX FTS – JEE ADV. FULL SYLLABUS TEST-04 [P1]

KEY SHEET

MATHEMATICS

1	**ABC**	2	**AD**	3	**BC**	4	**C**	5	**A**	6	**C**
7	**B**	8	**18**	9	**12**	10	**4**	11	**6**	12	**2**
13	**1**	14	**B**	15	**A**	16	**D**	17	**D**		

PHYSICS

18	**ABC**	19	**BCD**	20	**AB**	21	**C**	22	**C**	23	**B**
24	**C**	25	**10**	26	**1**	27	**7**	28	**1**	29	**3**
30	**2**	31	**A**	32	**B**	33	**A**	34	**C**		

CHEMISTRY

35	**BC**	36	**ABCD**	37	**C**	38	**C**	39	**B**	40	**C**
41	**C**	42	**96**	43	**9**	44	**6**	45	**3**	46	**2400**
47	**2**	48	**B**	49	**D**	50	**D**	51	**A**		

SOLUTIONS
MATHEMATICS

1. Let $a + c + 2b = x$ ……(1)
$a + b + 2c = y$ ……(2)
$a + b + 3c = z$ ……(3)
$c = z - y$; $b = x + z - 2y$
$a = -x + 5y - 3z$

$$\left(\frac{2y-x}{x}\right)+\frac{4(x+z-2y)}{y}-\frac{8(z-y)}{z} = -17+2\left(\frac{y}{x}+\frac{2x}{y}\right)+4\left(\frac{z}{y}+\frac{2y}{z}\right)$$

$$= -17 + 4\sqrt{2} + 8\sqrt{2} = 12\sqrt{2} - 17$$

2. $$f'(x) = 2\ell nx\frac{1}{x}+\frac{2}{x} = \frac{2}{x}(\ell nx+1)$$

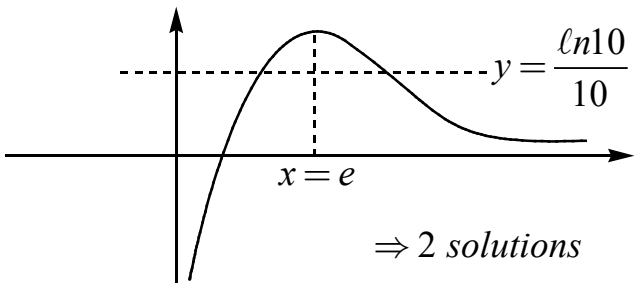
Also, slope of tangent = $\frac{f(x)}{x} \Rightarrow \frac{2}{x}(\ell nx+1) = \frac{(\ell nx)^2+2\ell nx}{x}$

$$\Rightarrow (\ell nx)^2 = 2 \Rightarrow x = e^{\sqrt{2}}, e^{-\sqrt{2}}$$

$$5x\,.\,f'(x) - x\ell n10 = 10 \Rightarrow 5x\frac{2}{x}(\ell nx+1) - x\ell n10 = 10$$

$$\Rightarrow 10\ell nx + 10 - x\ell n10 = 10 \Rightarrow \frac{\ell nx}{x} = \frac{\ell n10}{10}$$

Let $y = \frac{\ell nx}{x}$



$\Rightarrow$ 2 *solutions*

3. $\vec{b}\times\vec{d} = 0$
$\vec{c}\times\vec{d} = 0$

Take cross with $\vec{d}$ in equation $\vec{a} = \vec{b} + \vec{c}$

$$\Rightarrow \vec{a}\times\vec{d} = 0 + \vec{c}\times\vec{d} \Rightarrow \vec{d}\times\left(\vec{a}\times\vec{d}\right) = \vec{d}\times\left(\vec{c}\times\vec{d}\right)$$

$$= \left(\vec{d}\,.\,\vec{d}\right)\vec{c} - \left(\vec{d}\,.\,\vec{c}\right)\vec{d} = \left|\vec{d}\right|^2\vec{c} - 0 \Rightarrow \frac{\vec{d}\times\left(\vec{a}\times\vec{d}\right)}{\left|\vec{d}\right|^2} = \vec{c}$$

$$\Rightarrow \frac{\left|\vec{d}\times\left(\vec{a}\times\vec{d}\right)\right|}{\left|\vec{d}\right|^2} = \left|\vec{c}\right| = 2$$

As we have $\vec{a}=\vec{b}+\vec{c}\Rightarrow\vec{a}.\vec{d}=\vec{b}.\vec{d}+\vec{c}.\vec{d}$

$\Rightarrow\vec{a}.\vec{d}=\vec{b}.\vec{d}$ ……. (1)

Also as we know $\vec{b}\times\vec{d}=0$

Take cross with $\vec{a}$ both sides $\vec{a}\times\left(\vec{b}\times\vec{d}\right)=0\Rightarrow\left(\vec{a}.\vec{d}\right)\vec{b}-\left(\vec{a}.\vec{b}\right)\vec{d}=0$

$\Rightarrow\left(\vec{b}.\vec{d}\right)\vec{b}=\left(\vec{a}.\vec{b}\right)\vec{d}$ (from (1))

4. $f(x)=\left(x^2+\frac{1}{x^2}\right)+\left(x^4+\frac{1}{x^4}\right)+.....\left(x^{2n}+\frac{1}{x^{2n}}\right)+2n$

$$f(x)-2n=x^2\left(\frac{x^{2n}-1}{x^2-1}\right)+\frac{1}{x^2}\left(\frac{1-\frac{1}{x^{2n}}}{1-\frac{1}{x^2}}\right)=\left(\frac{x^{2n}-1}{x^2-1}\right).\left[x^2+\frac{1}{x^{2n}}\right]$$

$$g(x)=\lim_{n\to\infty}\left(\frac{1}{x^{2n+2}}.\left(1-x^{2n}\right).\left(x^2+\frac{1}{x^{2n}}\right)\right)$$

$$=\lim_{n\to\infty}\left[\left(\frac{1}{x^{2n}}-1\right).\left(1+\frac{1}{x^{2n+2}}\right)\right]=\begin{cases}-1 & |x|>1\\ 1 & x=\pm1\\ \infty & |x|<1\end{cases}$$

5. Let $I=\int_0^{\pi}\frac{x.dx}{1-\cos\phi\,.\sin x}$ ……. (1)

$$\Rightarrow I=\int_0^{\pi}\frac{\pi-x}{1-\cos\phi.\sin(\pi-x)}dx$$ ……. (2)

Adding (1) and (2) $\Rightarrow 2I=\pi\int_0^{\pi}\frac{dx}{1-\cos\phi.\sin x}$

$$I=\frac{\pi}{2}\int_0^{\pi}\frac{dx}{1-\cos\phi\frac{2\tan\frac{x}{2}}{1+\tan^2\frac{x}{2}}}=\frac{\pi}{2}\int_0^{\pi}\frac{\sec^2\frac{x}{2}dx}{1+\tan^2\frac{x}{2}-2\cos\phi\tan\frac{x}{2}}$$

Putting $\tan\frac{x}{2}=t\Rightarrow\frac{1}{2}\sec^2\frac{2x}{2}dx=dt$

$$I=\pi\int_0^{\infty}\frac{dt}{1+t^2-2t.\cos\phi}=\pi\int_0^{\infty}\frac{dt}{(t-\cos\phi)^2+\sin^2\phi}=\pi\left[\frac{2}{\sin\phi}.\tan^{-1}\left(\frac{t-\cos\phi}{\sin\phi}\right)\right]_0^{\infty}$$

$$I=\pi.\frac{1}{\sin\phi}\left[\frac{\pi}{2}-\left(\tan^{-1}(-\cot\phi)\right)\right]=\frac{\pi}{\sin\phi}\left[\frac{\pi}{2}+\frac{\pi}{2}-\phi\right]$$

$$= \frac{\pi.(\pi-\phi)}{\sin\phi} = f(\pi-\phi) \qquad \text{(As given)}$$

$$\Rightarrow f(x) = \pi\frac{x}{\sin x} \quad \text{To find } \int_0^{\frac{\pi}{2}} \pi^2 \frac{x^2}{\sin^2 x} = \pi^2 \int_0^{\frac{\pi}{2}} x^2.\cos ec^2 x$$

$$= \pi^2\left[\left(x^2(-\cot x)\right)_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} 2x.\cot x.dx\right] = \pi^2\left[0 + (2x.\ell n\sin x)_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} 2.\ell n\sin x.dx\right]$$

$$= \pi^2\left[0 - 2\left(-\frac{\pi}{2}\ell n2\right)\right] = \pi^3\ell n2$$

6. $\frac{d}{dx}g(x) = \frac{d}{dx}\frac{1}{x}\int_0^x f(t)\,dt$

$$\frac{d}{dx}g(x) = \frac{1}{x}f(x) + \int_0^x f(t)\,dt\left(-\frac{1}{x^2}\right) \quad \ldots\ldots\ldots(1)$$

and as g'(2) = 0, we must have $-\frac{1}{4}\int_0^2 f(t)\,dt + \frac{f(2)}{2} = 0 \Rightarrow \int_0^2 f(t)\,dt = 2f(2) = 10$

Now,

$$\frac{d^2}{dx^2}g(x) = \frac{d}{dx}\left(-\frac{1}{x^2}\int_0^x f(t)\,dt + \frac{1}{x}f(x)\right) = \frac{2}{x^3}\int_0^x f(t)\,dt - \frac{1}{x^2}f(x) - \frac{1}{x^2}f(x) + \frac{1}{x}f'(x)$$

$$\frac{d^2}{dx^2}g(2) = \frac{1}{4}\int_0^2 f(t)\,dt - \frac{f'(2)}{2} = -\frac{3}{2} < 0$$

Hence, g(x) has a local maximum at x = 2

7. $\frac{1}{x^2}\frac{dx}{dy} + \frac{y}{x} = \left(1+\frac{1}{y^2}\right)$

Let $\frac{1}{x} = t$ $\qquad \frac{dt}{dy} - yt = -\left(1+\frac{1}{y^2}\right), te^{-\frac{y^2}{2}} = -\int\left(1+\frac{1}{y^2}\right)e^{-\frac{y^2}{2}}dy$

$$\frac{1}{x} = \frac{1}{y} + ce^{\frac{y^2}{2}}, f(1,2) = 0 \Rightarrow c = \frac{1}{2}e^{-2}$$

$$\frac{2(y-x)}{xy} = e^{\left(\frac{y^2}{2}-2\right)}$$

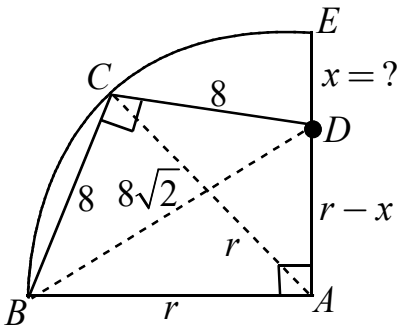
hyperbola $\frac{2(y-x)}{xy}=1$ & $\frac{2(y-x)}{xy}=e^{\left(\frac{y^2}{2}-2\right)}$ intersect where $y=\pm 2$ so point is (1, 2).

8. Mean is the same $\Rightarrow \frac{x^2+y^2-(x+1)^2-(y-1)^2}{5}=2$

$\Rightarrow -2x+2y-2=10$

$\Rightarrow y-x=6 \Rightarrow$ Minimum value of x^2+y^2 is $\left(3\sqrt{2}\right)^2=18$

9.



Ptolemy's theorem: (ABCD is cyclic)

$r.8\sqrt{2}=8r+8(r-x)$

$r\sqrt{2}=r+r-x \Rightarrow r-x=r\left(\sqrt{2}-1\right)$

$x=2r-r\sqrt{2}=r.\left(2-\sqrt{2}\right)$

In $\Delta ABD \Rightarrow r^2+(r-x)^2=\left(8\sqrt{2}\right)^2=128$

$r^2+r^2\left(\sqrt{2}-1\right)^2=128$

Find r, hence find x.

10. Put $\sin 2\theta = x$ given equation becomes $x^4+1=17\left(1+2x+x^2\right)^2$

Divide by x^2, we have $x^2+\frac{1}{x^2}=17\left(x+\frac{1}{x}+2\right)$

Now put $y=x+\frac{1}{x} \Rightarrow y=-\frac{7}{4}$ or $-\frac{5}{4}$

Now $y=-\frac{7}{4}$ have no solution and $y=-\frac{5}{4}$, we have $x=-\frac{1}{2},-2$

So, there are (4) solution in $[0,2\pi]$

11. $f\left(e^{-\frac{i\pi}{3}}\right)=(a+c)\omega-b+c;\omega=\frac{1+i\sqrt{3}}{2}$

$\Rightarrow (a+c)\left(-\frac{1}{2}+\frac{i\sqrt{3}}{2}\right)-b+c=4038\sqrt{3}i-4032$

$\Rightarrow a+c=8076; \left(\frac{c-a}{2}\right)-b=-4032 \Rightarrow a=c=4038$ and $b=4032$

12. $\left(e^{7x}-1\right)-\left(e^{7x}-1\right).e^{3x}-6.\left(e^{3x}\right)^2=0$

let $e^{7x}-1=t, e^{3x}=s$

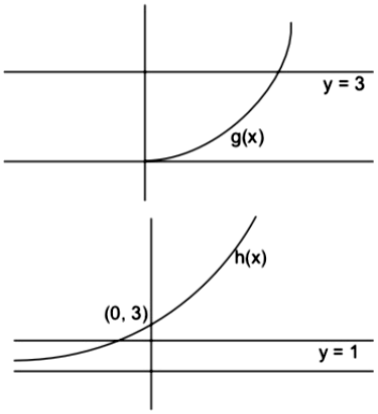
$t^2-ts-6s^2=0 \quad \Rightarrow (t-3s)(t+2s)=0$

$\Rightarrow t=3s$, or t + 2s = $0 \Rightarrow e^{7x}-1=3e^{3x}$ or $e^{7x}-1+2e^{3x}=0$

$e^{4x}-e^{-3x}=3$

Let $g(x)=e^{4x}-e^{-3x}$

$g'(x)>0$ only one solution



or $e^{7x}+2e^{3x}=1$

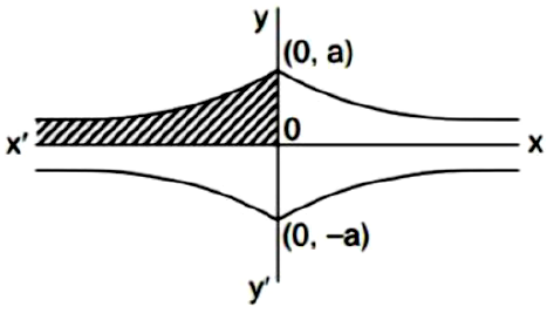
$h(x)=e^{7x}+2e^{3x}$

$h'(x)>0$

Only one solution

Hence total number of real solutions two

13. Required area, $A=4\int_0^{\pi/2} y.\frac{dx}{dt}dt$



$=4\int_0^{\pi/2}(a\sin t)\left(\frac{a}{\sin t}\cos^2 t\right)dt=4a^2\int_0^{\pi/2}\cos^2 t\,dt=\pi a^2$

14. **(A)** line $L\equiv y=x \Rightarrow$ image of P(2, 4) in L will be Q(4, 2)

Circle S: $(x-4)^2+(y-2)^2=16 \quad \Rightarrow x^2+y^2-8x-4y+4=0$

Circle M touches y = x at (1, 1)

$M:(x-1)^2+(y-1)^2+\lambda.(x-y)=0$

$\Rightarrow x^2+y^2+x(\lambda-2)-y(\lambda+2)+2=0$

Now radical axis of circles S, M : S – M = 0

$\Rightarrow -8x-x(\lambda-2)-4y+y(\lambda+2)+2=0$ passes through P(2, 4)

$\Rightarrow -16-2\lambda+4-16+4\lambda+8+2=0\Rightarrow 2\lambda=18\Rightarrow\lambda=9$

$M:x^2+y^2+7x-11y+2=0$

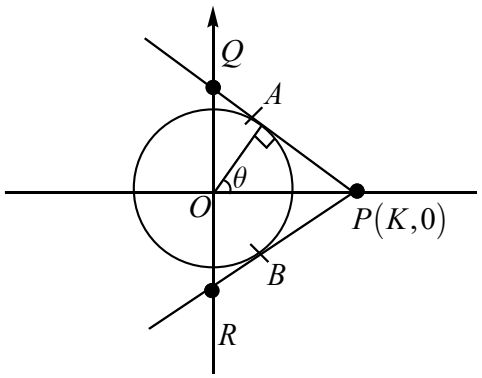
Radius = $\sqrt{\frac{49}{4}+\frac{121}{4}-2}=\frac{\sqrt{162}}{2}$

(B) Here $OP=K, OA=\sqrt{50}$

$K.\cos\theta=\sqrt{50}$

$K=\sqrt{50}.\sec\theta$

Also, $\angle OQP=\theta \quad \Rightarrow OQ=\sqrt{50}.\cos ec\theta$



Area of $\Delta PQR=\frac{1}{2}.OP.20Q=\sqrt{50}.\sec\theta.\sqrt{50}.\cos ec\theta=\frac{50}{\sin\theta.\cos\theta}$

$=\frac{100}{\sin 2\theta}\Rightarrow Min.\ area=100$

When $\theta=45^0 \qquad \Rightarrow K=\sqrt{50}\times\sec 45^\circ=\sqrt{50}.\sqrt{2}=10$

(C) $2x^2+xy+y^2=5\rightarrow$ center (0, 0)

Put $x=r.\cos\theta, y=r.\sin\theta\Rightarrow r^2=\frac{5}{2\cos^2\theta+\sin\theta.\cos\theta+\sin^2\theta}$

$=\frac{10}{2+2\cos^2\theta+2\sin\theta.\cos\theta}=\frac{10}{3+\cos 2\theta+\sin 2\theta}$

For length of major axis r is maximum.

$r^2_{\max}=\frac{10}{3-\sqrt{2}}\Rightarrow a=3, b=\sqrt{2}\Rightarrow a^2.b=9.\sqrt{2}=\sqrt{162}$

(D) Tangent to $\frac{x^2}{25}-\frac{y^2}{24}=1$ at $\left(5\sqrt{2},2\sqrt{6}\right)$ will be $\frac{x.5\sqrt{2}}{25}-\frac{y.2\sqrt{6}}{24}=1$

Intersecting tangent at vertices $x=\pm 5$ at points A, B

$\Rightarrow\pm\sqrt{2}-\frac{y}{2\sqrt{6}}=1\Rightarrow y=2\sqrt{6}\left(\sqrt{2}-1\right)$ or $2\sqrt{6}\left(\sqrt{2}+1\right)$

Now segment of any tangent between tangent at vertices subtends 90° at foci, circle drawn through these 4 points will have points A, B as diameter.

$$\Rightarrow K=\frac{2\sqrt{6}\left(\sqrt{2}-1\right)-2\sqrt{6}\left(\sqrt{2}+1\right)}{2}=-2\sqrt{6}\Rightarrow\left|2\sqrt{3}.K\right|=12.\sqrt{2}$$

15. $\dfrac{\left({}^8C_1+{}^8C_3+.....+{}^8C_7\right)^7\times 1}{2^{64}}$

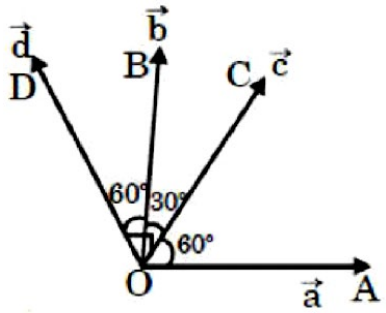
16. **(P)** $\left|A_{4\times 3}\ B_{3\times 4}\right|$ is 0

(Q) $|A|=0$ and $|B|=0$

(R) $AB^T=B^TA$

(S) $\left|2A^2B^{-3}\right|=8\times 1^2\times\dfrac{1}{27}$

17.



($\hat{p}$ is unit vector $\perp$ to the plane)

$\sin\theta_1\sin\theta_2\cos\phi=1,\ \cos\alpha=\dfrac{1}{2}$

$\theta_1=\theta_2=90^\circ,\ \phi=0,\ \alpha=\dfrac{\pi}{3}\Rightarrow$ OABCD are coplanar

$$\Delta ABC=\frac{1}{2}\left|\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}\right|\ =\frac{1}{2}\left|2\hat{p}-3\hat{p}-\frac{3\sqrt{3}}{2}\hat{p}\right|=\frac{1}{2}\left(1+\frac{3\sqrt{3}}{2}\right)$$

PHYSICS

18. Velocity of ball A along and perpendicular to line of impact after collision $\left(0,10\sin 30^\circ\right)$

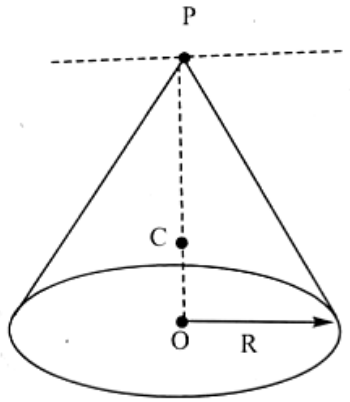
Velocity of ball B along and perpendicular to line of impact before collision (0,0)

Velocity of ball B along and perpendicular to line of impact after collision $\left(10\cos 30^\circ,0\right)$

19. $t=\dfrac{\ell}{v}+\dfrac{v(\alpha+\beta)}{2\alpha\beta}$ For to be minimum $\dfrac{dt}{dv}=0$

20. If the charge Q is placed at B, induced charge will be distributed uniformly on the outer surface of conducting sphere.

21.



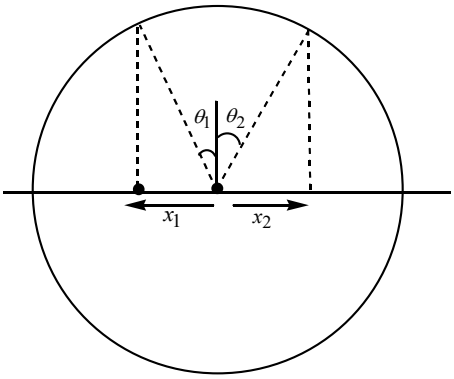
$$I_P = \frac{3}{5}m\left\{\frac{R^2}{4} + h^2\right\}$$

$$I_O = \frac{3}{5}m\left\{\frac{R^2}{4} + h^2\right\} - m\left(\frac{3h}{4}\right)^2 + m\left(\frac{h}{4}\right)^2 = \frac{3}{20}mR^2 + \frac{1}{10}mh^2$$

22. Result must contain one significant figures

23. $v = \sqrt{gR}; \qquad \omega = \sqrt{\frac{g}{R}}; \qquad x = \frac{r}{2}$

$$v^2 = \omega^2\left(A^2 - x^2\right) \Rightarrow \cancel{g}R = \frac{\cancel{g}}{R}\left(A^2 - \frac{R^2}{4}\right) \Rightarrow \qquad A = \frac{\sqrt{5}}{2}R$$



$$t = \frac{\theta_1 + \theta_2}{\omega} = \left\{\sin^{-1}\left(\frac{\frac{R}{2}}{\sqrt{5}\frac{R}{2}}\right) + \sin^{-1}\left(\frac{R}{\sqrt{5}R/2}\right)\right\}\sqrt{\frac{R}{g}} = \left\{\sin^{-1}\left(\frac{1}{\sqrt{5}}\right) + \sin^{-1}\left(\frac{2}{\sqrt{5}}\right)\right\}\sqrt{\frac{R}{g}}$$

24. $U_B = \frac{1}{2}Li^2$

$$U_E = \frac{1}{2}\frac{q^2}{C}$$

25. $S_{rel} = 5m$

$$T = \frac{2u_y}{g} = \frac{u_y}{5} \qquad 40 \times v_1 + 40v_2 = 80 \times 8$$

$v_1 + v_2 = 16 m/s \Rightarrow v_2 = 16 - v_1$

$v_1 - 8 = \frac{5}{T} = \frac{25}{u_y} \Rightarrow v_1 = 8 + \frac{25}{u_y}$

$v_1 - v_2 = v_{rel}x = 2v_1 - 16 = 16 + \frac{50}{u_y} - 16$

$v_{rel}y = u_y$

$v_{rel} = \sqrt{\frac{50^2}{u_y^2} + u_y^2 v_{rel_{\min}} xy_y - \frac{50^2}{u_y^3}} \times 2 = 0$

$u_y^4 = 50^2 \qquad u_y = \sqrt{50} = 5\sqrt{2}$

$v_{rel} = \sqrt{50+50} = 10 m/s$

26. If potential at the corner of solid cube is V

$V \propto \frac{m}{\ell}; m \propto \ell^3 \Rightarrow V \propto \ell^2$ (ℓ is side length of cube)

Potential at centre of cube $= \left(\frac{V}{4}\right) \times 8 = 2V$

Work done in slowly moving a point mass from centre to corner =

$= m\left(V_{corner} - V_{centre}\right) = m\left\{V - (2V)\right\} = -mV = W$

Work done in slowly moving same particle from corner to infinity =

$= m\left(V_{\inf inity} - V_{corner}\right) = m(O - V) = -mV = W$

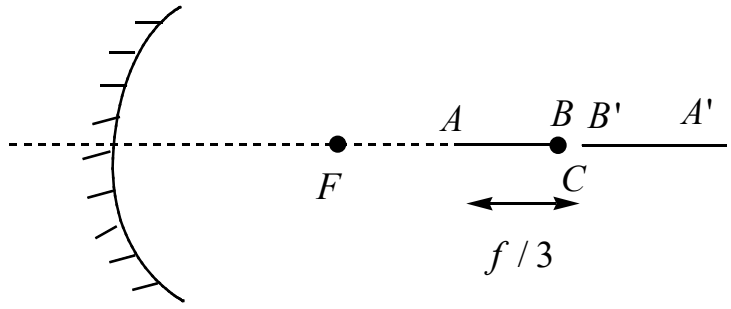
$W' = W$

27. $\overline{\nabla}.\overline{E} = \frac{1}{r^2}\frac{\partial}{\partial r}\left\{r^2.kr^3\right\} = \frac{1}{r^2} \times 5kr^4 = 5kr^2 \quad \because \overline{\nabla}.\overline{E} = \frac{\rho}{\varepsilon_0} \Rightarrow \rho = 5k\varepsilon_0 r^2$

28. Speed of transverse wave $= V_t = \sqrt{\frac{T}{\rho A}}$

Speed of longitudinal wave $= V_t = \sqrt{\frac{T}{\rho}}$

29.



For image of A

$u = -\frac{5f}{3} \qquad \Rightarrow v = -\frac{5f}{2}$

$\therefore$ length of $A'B' = \frac{f}{2} \quad \therefore$ magnification $= \frac{A'B'}{AB} = \frac{3}{2}$

30. $\omega^2 A = \mu g \Rightarrow A = \frac{\mu g}{4\pi^2 f^2} = 0.02m$

31. P) $\tan\theta = \frac{1}{T\alpha} \Rightarrow 3 = \frac{VdT}{TdV} \Rightarrow \frac{3dV}{V} = \frac{dT}{T}$

$\ln V^3 = \ln T + C \Rightarrow TV^{-3} =$ constant

This is polytropic process with $x - 1 = -3 \Rightarrow x = -2$

$$W = \frac{nR}{x-1}(T_1 - T_2) = \frac{2\left(\frac{25}{3}\right)}{-3}(-1) = +\frac{50}{9}J$$

Q) $\Delta U = nC_V \Delta T = \frac{2\left(\frac{25}{3}\right)}{\left(\frac{5}{3} - 1\right)}(1) = 25J$

R) $\Delta Q_P = nC_P \Delta T = (2)\left[\frac{5R}{2}\right]\Delta T = 5 \times \frac{25}{3}(1) = \frac{125}{3}J$

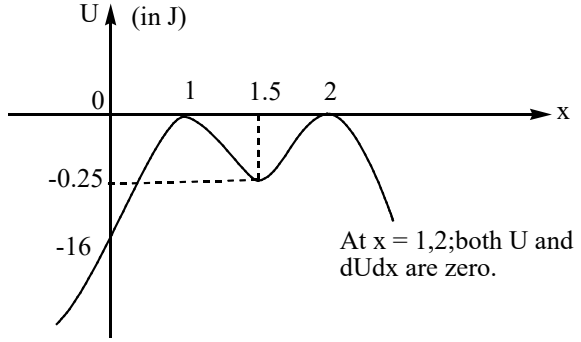
S) $\Delta U = 25J$ for any polytropic co-efficient i.e., for any value of 'θ'.

32. $\frac{dU}{dx} = -4\left[2(x-1)(x-2)^2 + 2(x-1)^2(x-2)\right]$

Or, $\frac{dU}{dx} = -8(x-1)(x-2)[x-2+x-1] = -8(x-1)(x-2)(2x-3)$

Force on the particle is $F = -\frac{dU}{dx} = 8(x-1)(x-2)(2x-3)$ (ii)

Force $\left(i.e., \frac{dU}{dx}\right)$ is zero at x = 1,2,1.5m. These are positions of equilibrium. Nature of the equilibrium can be ascertained by finding $\frac{d^2U}{dx^2}$ or just by plotting the graph of U versus x.



A) From (ii), the force on the particle is negative at x = –1m. If it is released at x = –1m, it will move in the negative x direction.
B) If the particle is released at x = 1m, it will remain in unstable equilibrium.
C) Force F = 8 (x – 1) (x – 2) (x – 3)
When the particle is displaced Δx from its equilibrium position at x = 1.5m (so that its co-ordinate is $1.5+\Delta x$), the force on it is

$$F = 8(1.5+\Delta x-1)(1.5+\Delta x-2)(3+2\Delta x-3)$$

$$= 8(0.5+\Delta x)(\Delta x-0.5)(2\Delta x) = 8\left(\Delta x^2-0.5^2\right)(2\Delta x) = 16\Delta x\left(\Delta x^2-0.5^2\right)$$

If $\left(\Delta x^2-0.5^2\right)(2\Delta x) = 16\Delta x\left(\Delta x^2-0.5^2\right)$

Or, $F = 16\Delta x\left(-0.5^2\right) = -4\Delta x \therefore\ ma = -4\Delta x \Rightarrow a = -4\Delta x \quad [\because m = 1kg]$

$\therefore$ Motion is SHM and $\omega = \sqrt{4} = 2$
D) It is same as (C)

33. $\overline{F} = q\left\{\overline{E}+\overline{\upsilon}\times\overline{B}\right\}$

34. $E_2 = \dfrac{E_1}{L}\times\ell$ where L = length of AB
A) If E_1 is increased, ℓ should be decreased keeping E_2 and L constant.
B) If R is increased, current through potentiometer will decrease. Hence V_{AJ} will decrease. But $V_{AJ} = E_2$ means length should be increased.
C) If E_2 is increased, 1 should be increased.

CHEMISTRY

35. 1) Copper pyrite $\left(CuFeS_2\right)$ upon roasting (heating in the presence of air) produces a mixture of Cu_2S and FeS, which upon further smelting and refining yields blister copper (containing about 98% Cu and impurities like Fe, Ni, Zn, Au, Ag).
2) Magnesite $\left(MgCO_3\right)$ upon calcination (heating in the absence of air) yields magnesia (MgO): $MgCO_3 \rightarrow MgO + CO_2$
3) Nickel is purified by the Mond process. Impure nickel is reacted with CO at low temperature and pressure to form volatile nickel carbonyl $\left[Ni(CO)_4\right]$. This is then decomposed at a higher temperature to give pure nickel:
$Ni + 4CO \rightleftharpoons Ni(CO)_4$ (low temp/high pressure) $Ni(CO)_4 \rightarrow Ni + 4CO$ (high temp)
4) Bauxite $Al_2O_3xH_2O$ is the principal ore of aluminum. It is not directly heated with coke. Instead, alumina $\left(Al_2O_3\right)$ is extracted from bauxite and then electrolytically reduced to aluminum in the Hall-Héroult process. Coke (carbon) is used in the electrolytic reduction as a reducing agent in the form of graphite electrodes.

36. 1. $MeMgCl + H_2C = C = O \rightarrow CH_3COCH_3$ (Acetone)

2. $CH_3COCH_3 + H_2O \rightarrow CH_3COCH_3$ (Acetone) (workup step, no change in the product)

3. $CH_3COCH_3 + conc.H_2SO_4 \rightarrow$ Phorone

37. 1) Toluene to A $\left(Br_2 / hv\right)$: Bromination of toluene under free radical conditions (light or heat) occurs at the benzylic position, forming benzyl bromide $\left(PhCH_2Br\right)$.

2) A to B (alcoholic KOH): Benzyl bromide (A) undergoes an elimination reaction $\left(E_2\right)$ with alcoholic KOH. This strong base favors the formation of the more substituted alkene, leading to (o-tolyl)ethene ($PhCH = CHCH_3$ where the CH_3 group is ortho to the ethene group).

3) B to C $\left(O_3 / Zn / AcOH\right)$: Ozonolysis of (o-tolyl)ethene followed by a reductive workup (Zn/AcOH) cleaves the double bond. The carbon with the phenyl group forms benzaldehyde (PhCHO), while the other carbon with the methyl group forms acetaldehyde.

4) C to D $\left(PhMgBr / H_3O^+\right)$: Benzaldehyde (C) reacts with phenylmagnesium bromide (PhMgBr). The Grignard reagent adds to the carbonyl group, and subsequent hydrolysis $\left(H_3O^+\right)$ yields a tertiary alcohol, triphenylcarbinol $\left(Ph_3COH\right)$.

38. Chromium(III) ions $\left(Cr^{3+}\right)$, which are green in solution, can be oxidized to chromate ions $\left(CrO_4^{2-}\right)$, which are yellow, under alkaline conditions using hydrogen peroxide $\left(H_2O_2\right)$. The reaction is:

$2Cr^{3+} + 3H_2O_2 + 10OH^- \rightarrow 2CrO_4^{2-} + 8H_2O$

Acidic conditions are *not* favorable for the formation of chromate. In acidic solutions, dichromate $\left(Cr_2O_7^{2-}\right)$ is the predominant chromium species, and the equilibrium shifts away from chromate.

39. Option (b) accurately describes the standard and most reliable method. Kohlrausch's Law is crucial for obtaining $\Lambda°m$ for weak electrolytes.

40. $\therefore pH = \frac{1}{2}\left(P^{K_{a1}} + P^{K_{a2}}\right) = \frac{1}{2}\left(2.4 + 9.8\right) = 6.1$

41. $D - glu\cos e \xrightarrow{\text{Re}agent} D - gluconic\,acid$ is carried by Fehling solution

42. 1) Monomer Units per Chain: A linear silicone polymer with 12 silicon atoms requires 12 dichlorodimethylsilane monomers.

2) Theoretical Moles of Monomer (per mole of polymer): To theoretically produce 1 mole of 12-silicon-atom polymer chains, the chemist needs 12 moles of dichlorodimethylsilane.

3) Total Theoretical Moles of Monomer: The chemist wants to *actually* obtain 6 moles of the polymer. Therefore, they need to theoretically produce 6 * 12 = 72 moles of polymer.

4) Adjusting for Percent Yield: The reaction has a 75% yield. This means that only 75% of the starting material is converted into the desired product. To calculate the required starting amount, we use:

Theoretical Yield = (Actual Yield * 100) / Percent Yield

Theoretical yield of monomer = (72 moles * 100) / 75 = 96 moles.

Therefore, the chemist must initially react 96 moles of dichlorodimethylsilane to obtain 6 moles of the desired average polymer chains.

43. $Z = \dfrac{PV_m}{RT} = \dfrac{V_m}{V_m - b} = \dfrac{10b}{10b - b} = \dfrac{10}{9}$

44. $\Delta H = E_{a_f} - E_{a_b} \quad \Rightarrow -2 = E_{a_f} - 8 \Rightarrow E_{a_f} = 6k\,cal/mol$

Now, the fraction of molecules crossing energy barrier $= e^{-E_a/RT} = e^{-\frac{6000}{1000}}$

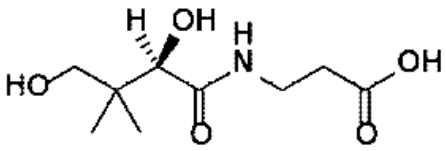
45. $q = 0 \Rightarrow \Delta U = w = -P_0\left(4V_0 - V_0\right) = -3P_0V_0$

Now, $\Delta H = \Delta U + \Delta\left(PV\right) = \left(-3P_0V_0\right) + \left(P_0 4V_0 - 2P_0V_0\right) = -P_0V_0$

46. $\Delta G = \Delta G^o + RT.\ln Q = -RT.\ln\dfrac{K_f}{K_b} + RT.\ln\dfrac{[\text{Product}]}{[\text{Reactants}]}$

$= RT.\ln\dfrac{K_b[\text{Product}]}{K_f[\text{Reactants}]} = RT.\ln\dfrac{r_b}{r_f}$

$= 2 \times 300 \times \ln\dfrac{1}{e^4} = -2400 \text{ cal}$

47. 

48. $CaH_2 \xrightarrow{H_2O} Ca\left(OH\right)_2 + H_2$

$POCl_3 \xrightarrow{H_2O} H_3PO_4 + HCl$

$NCl_3 \xrightarrow{H_2O} NH_3 + HOCl$

$B_2H_6 \xrightarrow{H_2O} H_3BO_3 + H_2$

$R_2SiCl_2 \xrightarrow{H_2O} R_2Si\left(OH\right)_2 + HCl$

49. A) $FeSO_4$ (heated)

2) $FeSO_4$ on heating decomposes to give Fe_2O_3 (Ferric Oxide), SO_2 (Sulfur Dioxide), and SO_3 (Sulfur Trioxide).

3) Fe_2O_3 is a reddish-brown solid.

4) SO_2 and SO_3 are gases. SO_3 is an acidic oxide.

5) The reaction is a redox reaction as the oxidation state of Iron changes.

Matches:

R) Redox reaction

S) One of the products is acidic oxide

B) $Mn^{2+} + S_2O_8^{2-} + H_2O$

1) This is a reaction between Manganese ions (Mn^{2+}) and Peroxodisulfate ions $\left(S_2O_8^{2-}\right)$ in the presence of water.

2) $S_2O_8^{2-}$ is a strong oxidizing agent and will oxidize Mn^{2+} to a higher oxidation state.

3) The products will depend on the exact conditions, but generally, Mn^{2+} is oxidized to MnO_4^- (Permanganate ion).

4) MnO_4^- has Mn in the +VII oxidation state.

5) $S_2O_8^{2-}$ is reduced to SO_4^{2-} (Sulfate ion), where Sulfur is in the +VI oxidation state.

Matches:

P) One of the products is coloured due to charge transfer (MnO_4^- is purple)

Q) One of the products in +VI oxidation state (Sulfur in SO_4^{2-})

R) Redox reaction

C) $Na_2Cr_2O_7$ (saturated salt) + H_2SO_4 (conc.)

1) This is the reaction of Sodium Dichromate $\left(Na_2Cr_2O_7\right)$ with concentrated Sulfuric Acid $\left(H_2SO_4\right)$.

2) The reaction produces Chromic Acid $\left(H_2CrO_4\right)$, Sodium Sulfate $\left(H_2SO_4\right)$, and water.

3) Chromic Acid $\left(H_2CrO_4\right)$ is unstable and will further react to form Chromium Trioxide $\left(CrO_3\right)$, which is an acidic oxide.

4) CrO_3 is a strong oxidizing agent.

Matches:

P) One of the products is coloured due to charge transfer (CrO_3 is a red-orange solid)

Q) One of the products in +VI oxidation state (Cr in CrO_3)

S) One of the products is acidic oxide

D) $N_2H_4 + CuSO_4$

1) This is a reaction between Hydrazine $\left(N_2H_4\right)$ and Copper Sulfate $\left(CuSO_4\right)$.

2) $CuSO_4$ reacts with excess N_2H_4 to form a complex, $\left[Cu\left(N_2H_4\right)_2\right]SO_4$, which is a deep blue or violet solution.

3) The color is due to the charge transfer between Copper and the Hydrazine ligand.

4) The reaction is also a redox reaction.

Matches:

P) One of the products is coloured due to charge transfer

Q) One of the products in +VI oxidation state (Sulfur in SO_4^{2-})

R) Redox reaction

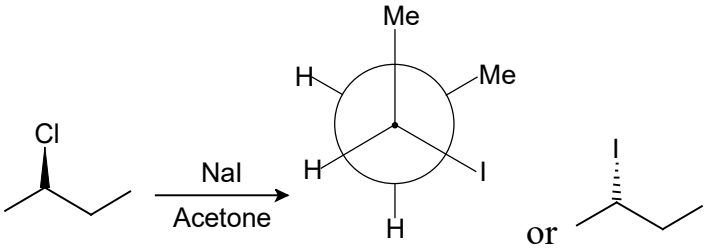
A) $FeSO_4$: R, S

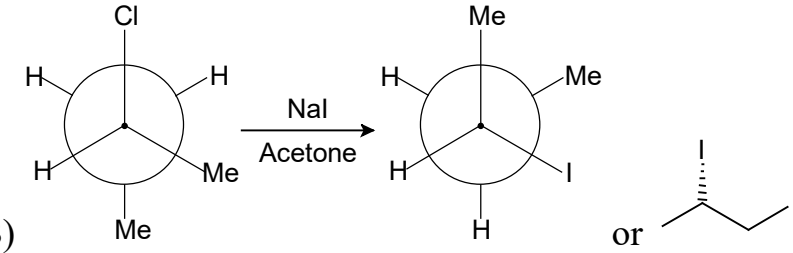
B) $Mn^{2+} + S_2O_8^{2-} + H_2O$: P, Q, R

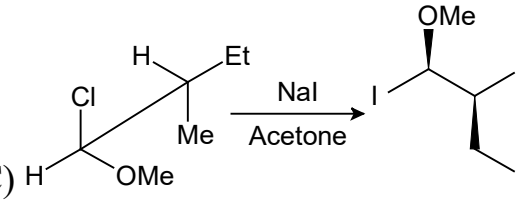
C) $Na_2Cr_2O_7 + H_2SO_4$: P, Q, S

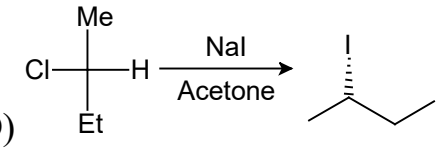
D) $N_2H_4 + CuSO_4$: P, Q, R

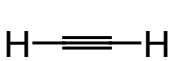
50. All are nucleophilic substitutions by second order(a single step reaction)

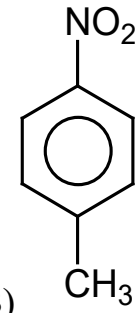
A) 

B) 

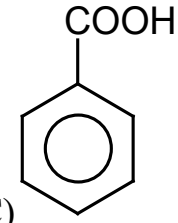
C) 

D) 

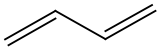
51. A)  $\xrightarrow[\text{(iii) Reduction with Ni/H}_2\ \text{(iv) NaNO}_2\text{+HCl}\ \text{(v) H}_2\text{O, Boil}]{\text{(i) Red hot iron, 873 K (ii) Fuming HNO}_3\text{, H}_2\text{SO}_4/\Delta} C_6H_6O$



B) $\xrightarrow[\text{(iii) NaOH+CaO,}\Delta]{\text{(i) KMnO}_4\ \text{OH}^-,\Delta\ \text{(ii) H}^+,\text{H}_2\text{O}} C_6H_5NO_2$



C) $\xrightarrow[\text{(iii) conc. H}_2\text{SO}_4\text{+conc. HNO}_3/\Delta]{\text{(i) NaOH+CaO/}\Delta\ \text{(ii) CH}_3\text{Cl(1 eq.)/AlCl}_3} C_7H_7NO_2$

D)  $\xrightarrow[\text{(iii) Cl}_2\text{/Fe (iv) CH}_3\text{Cl; AlCl}_3]{\text{(i) CH=CH, 200}^\circ\text{C (ii) Se/}\Delta} C_7H_7Cl$